



计算机系统结构

第一章

吴 非

华中科技大学信息存储系统教育部重点实验室
武汉光电国家实验室存储与显示功能实验室



计算机系统结构（微助教签到）



- 微助教签到

<https://www.teachermate.com.cn/classes/1468440>

课堂名称：计算机系统结构

课堂编号：RG889

- 理解计算机系统性能评价方法和工具





教师介绍



吴非

Mobile: 15071339557

Email: wufei@mail.hust.edu.cn

Office: WNLO B524

Homepage: nnsslabs.com

研究方向: 智能存储、存储智能、新型存储技术



曹强 Tel: 13277932288

Email: caoqiang@mail.hust.edu.cn

Office: WNLO B520

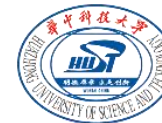
研究方向: 分布式存储、新型存储技术、光存储



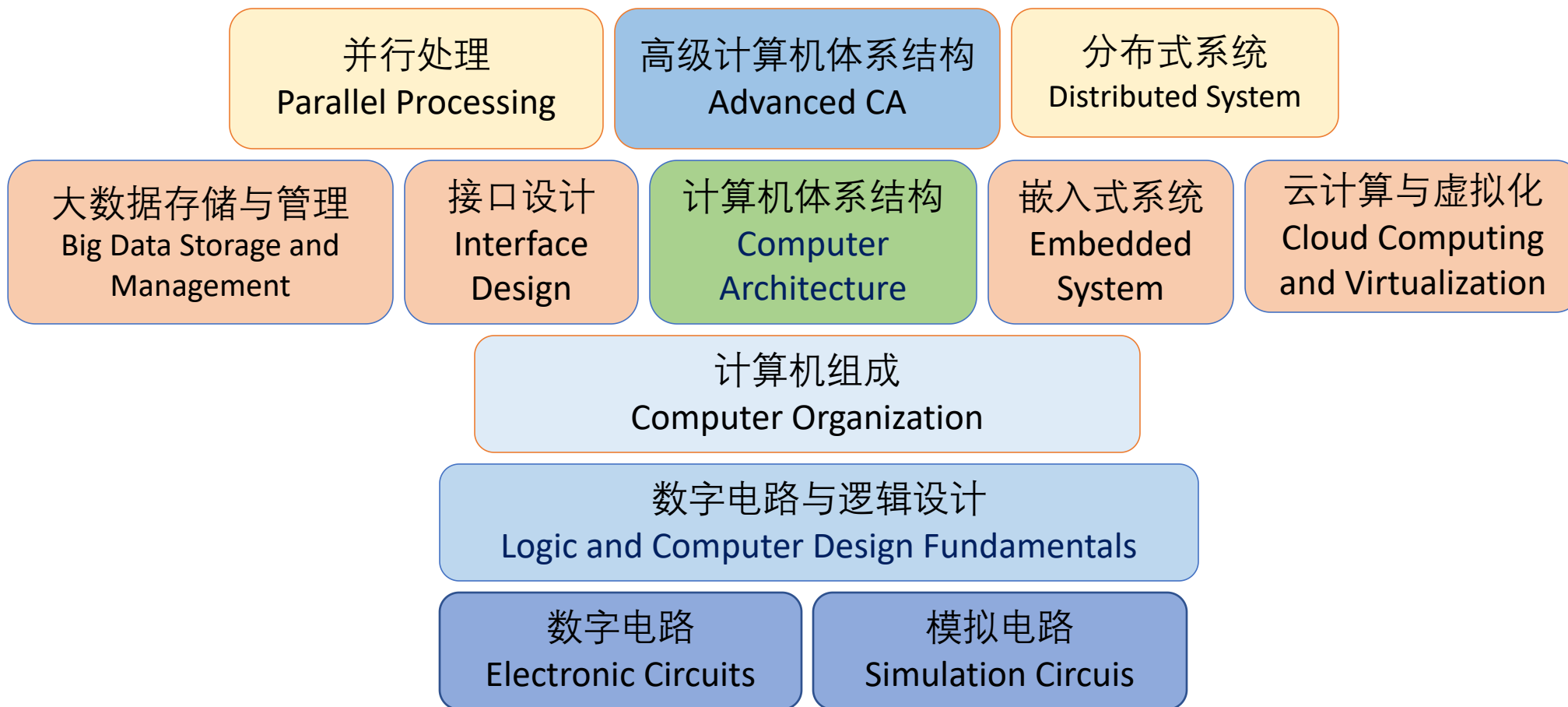
[illegible]



为什么学习计算机系统结构课程?

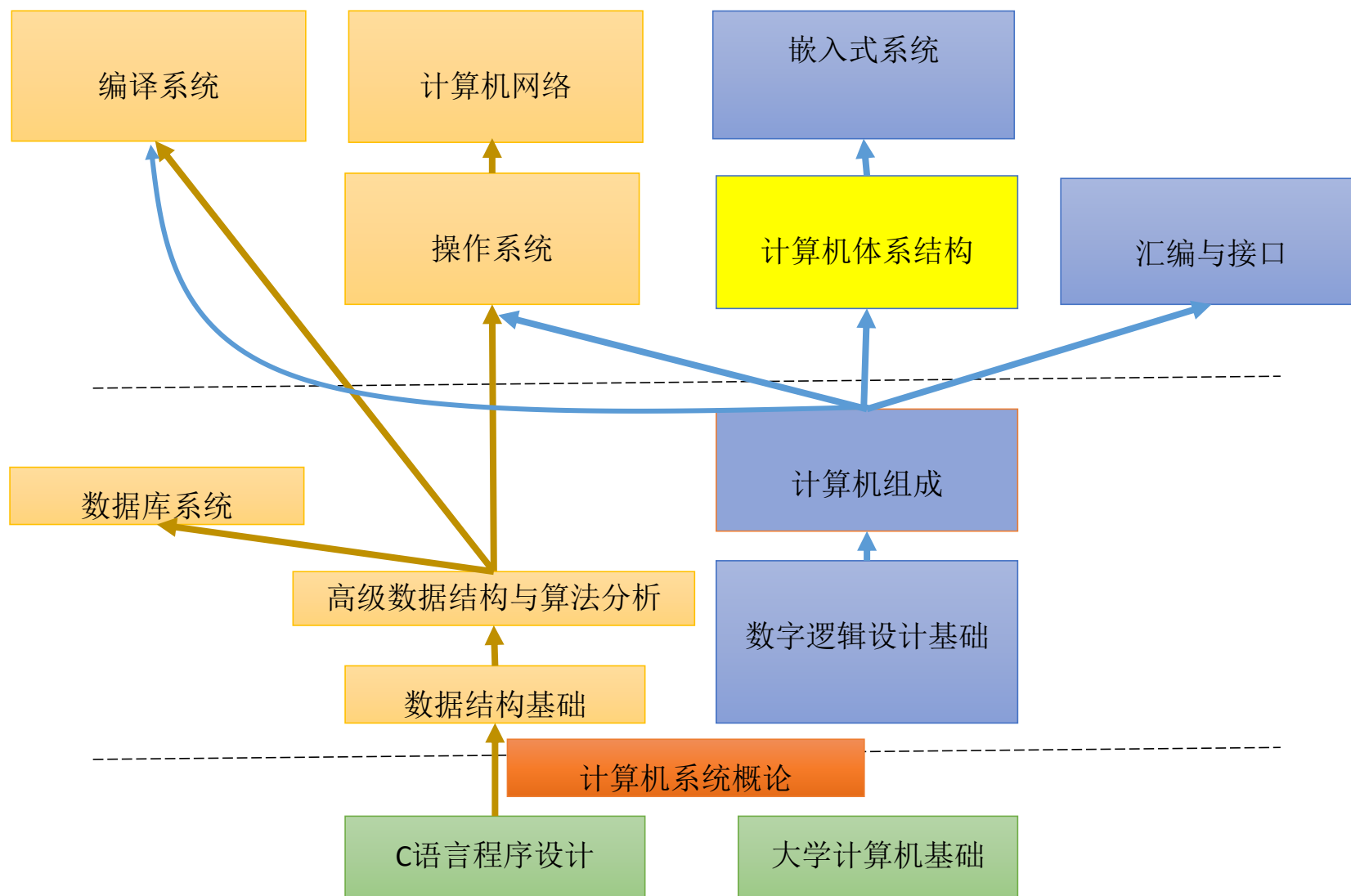


■ Hardware Course Series





为什么学习计算机系统结构课程?





课程间关系和区别



《计算机系统基础》

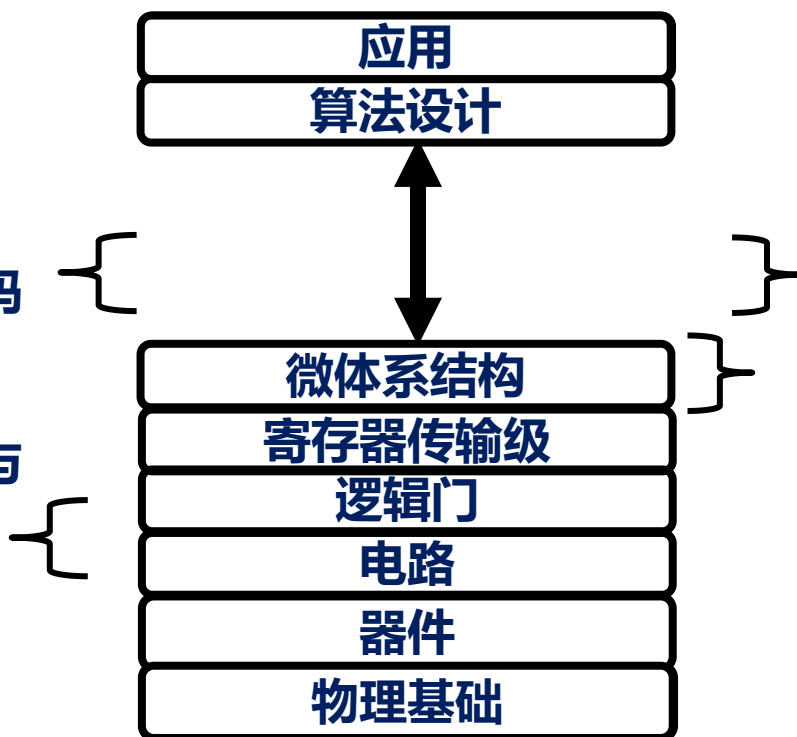
(从软件/硬件协同的角度)

- 能够选择合适的计算架构
- 写出效率和质量更高的代码
- 熟悉多种并行程序的特点

《计算机组成原理》 《数字系统结构与 设计》

(数字电路设计角度)

- 逻辑门-电路-功能基本知识
- 时序逻辑、组合逻辑设计
- 单周期/多周期流水线 (静态)



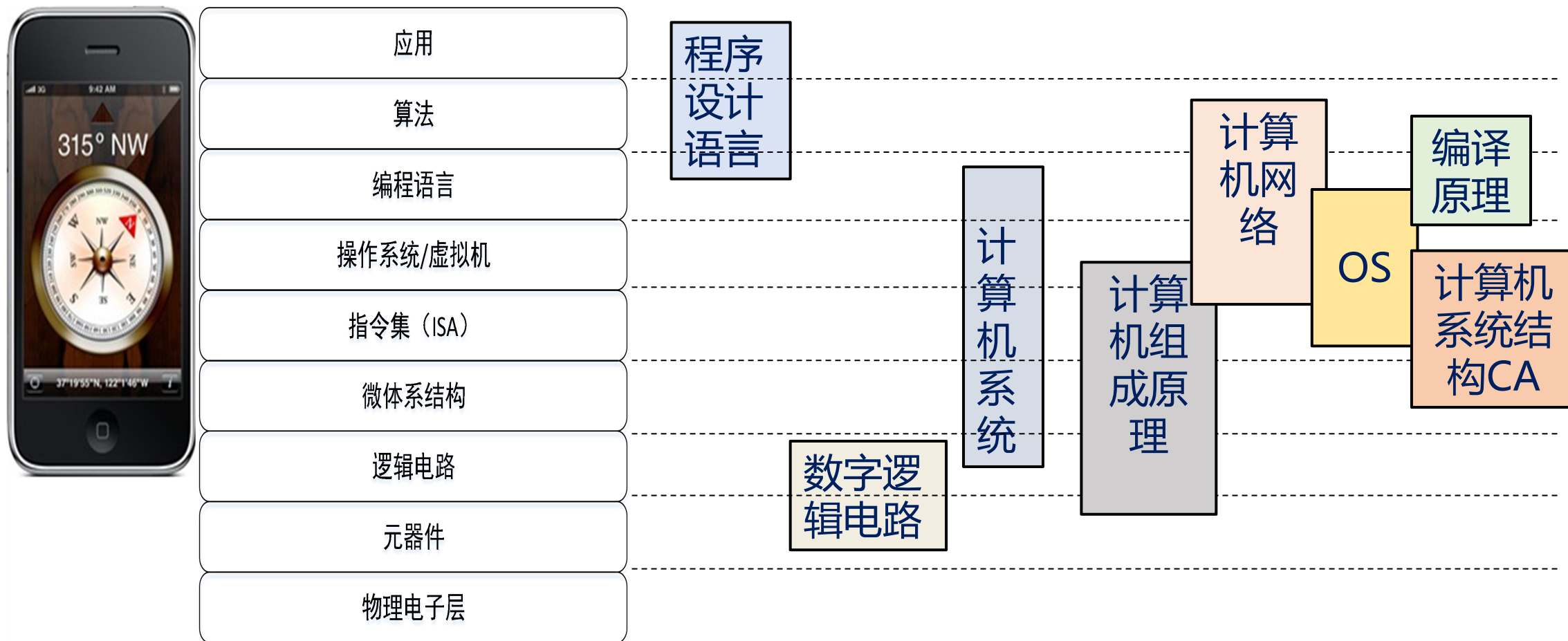
《计算机体系结构》

(从处理器设计者的角度)

- 理解指令、数据的抽象关系
- 掌握计算、存储和通讯内涵
- 熟悉不同计算架构的优缺点



计算机系统结构课程与其他课程的关系



CA讲述不同并行粒度的体系结构及其与其相关的OS实现和编译器的实现

课程教材



Instructor: Wu Fei & Cao Qiang

Lectures: Two 90-minutes lectures a week

Textbook:

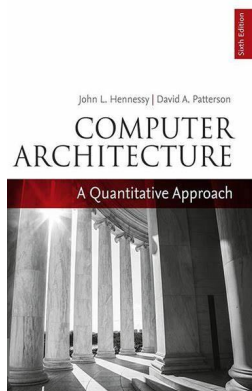
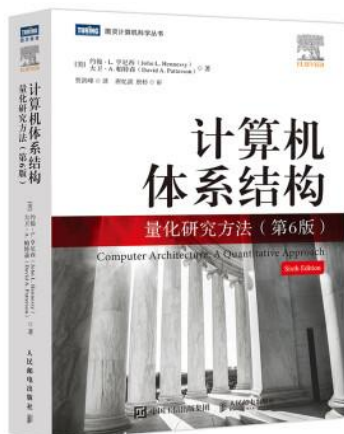
计算机系统结构——量化研究方法（第六版），John L.Hennessy, David A.Patterson，机械工业出版社（2022）

计算机系统结构（第四版），张晨曦，沈立、王志英等，清华大学出版社（2023）

计算机系统结构（微课版），曹强，施展，人民邮电出版社（2024）

Prerequisite:

计算机组成原理（微课版、第二版），谭志虎、秦磊华、吴非、周健、周游，人民邮电出版社（2025）





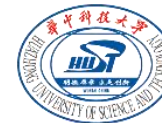
Difference of 6th & 5th edition



- Most significant change:
 - ISA selection
 - 5th MIPS
 - 6th RISC-V
 - Add a new chapter:
 - Chapter 7 Domain-Specific Architecture(DSA)
 - CPU/GPU/DNN accelerator comparison
- Other change:
 - Remove description of Delayed Branch
 - RISC-V removes it because the performance of branch predictor is powerful enough.
- Book size limitation:
 - More contents move to Appendix and CD



版本与技术变迁



版本	核心并行 / 架构	关键新增	应用驱动
1-3 版	ILP (指令级)	超标量、乱序、分支预测	单线程通用计算
第 4 版	TLP/DLP (线程 / 数据)	多核、SMT、SIMD、GPU 初现	多线程、多媒体、科学计算
第 5 版	TLP+DLP + 异构 + WSC	GPU 架构、移动计算、WSC、虚拟化	移动互联网、云计算、大数据、GPU 并行
第 6 版	TLP+DLP+DSA (领域专用)	DSA、TPU/NPU、AI 加速、软件 ↓ 司	深度学习、AI、图计算、专用加速



John L. Hennessy (Stanford)



- Former President of Stanford University during 2000 – 2016 (17 billion)
- Current Alphabet Chairman
- "Godfather of Silicon Valley, "
- In 1981, Hennessy initiated a project at Stanford that focused on a simpler computer architecture known as RISC. During a sabbatical leave in 1984-85 he cofounded MIPS Computer Systems, now known as MIPS Technologies, which specializes in the production of microprocessors SPARC.
- Received [Eckert-Mauchly Award](#) in 2001
- Received [Turing Award](#) in 2017



<https://engineering.stanford.edu/people/john-hennessy>



David A. Patterson (UC Berkeley)



- UC Berkeley (1976 – 2016)
- Currently Google TPU
- He led the design and implementation of **RISC** I (the foundation of the SPARC architecture)
- Inventor of **RAID**
- involved in the Network of Workstations (NOW) project
- Research Accelerator for Multiple Processors (RAMP)
- Received ACM Eckert-Mauchly Award in ISCA 2008
- Received Turing Award in 2017



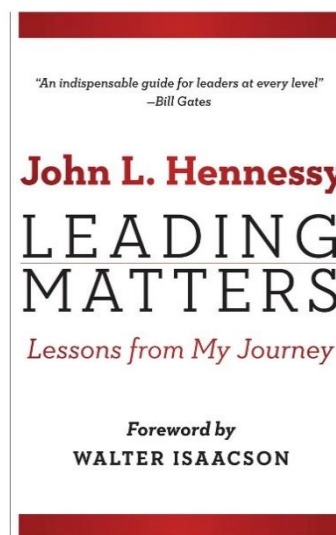
<https://people.eecs.berkeley.edu/~pattsrn/>



2018 interview



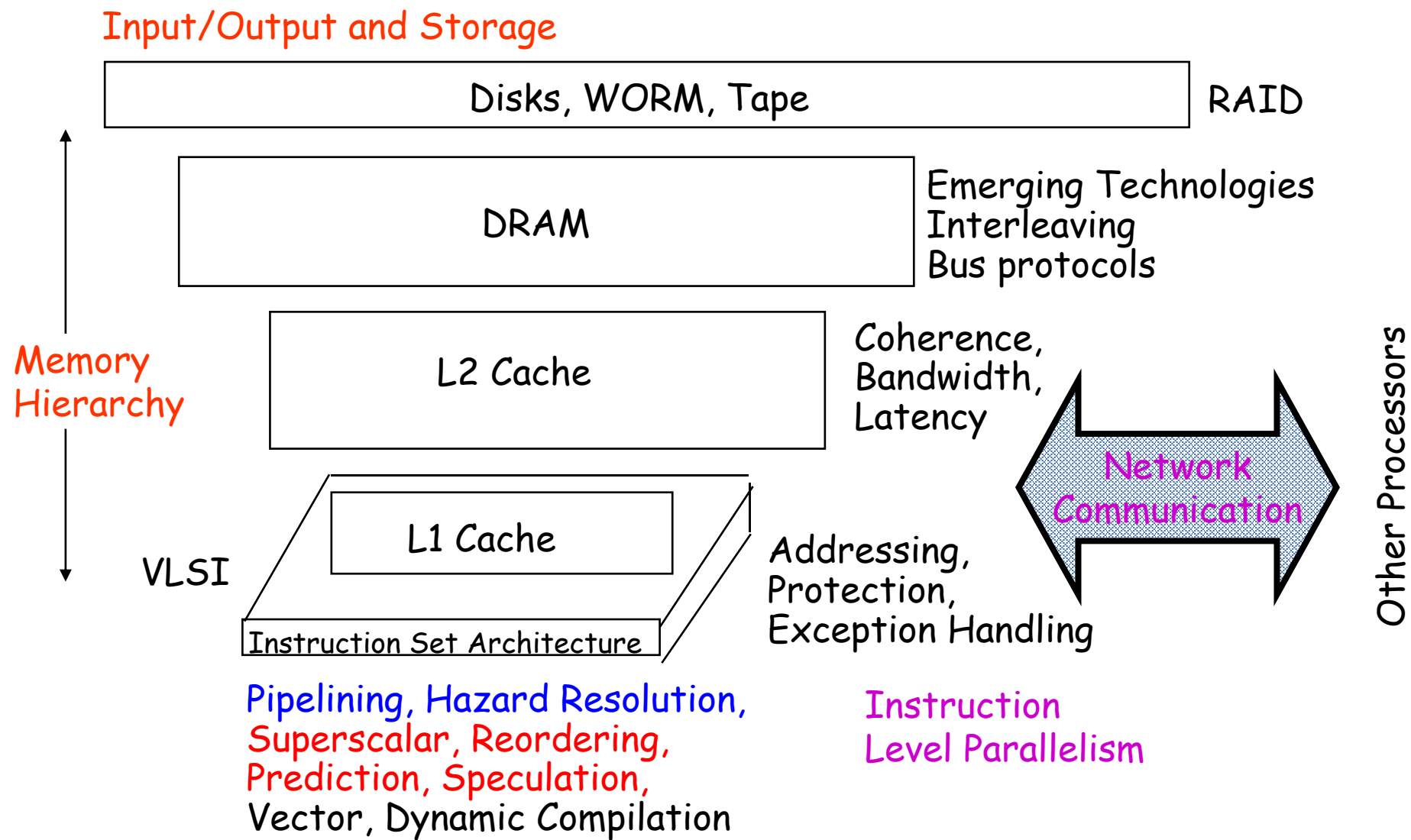
For pioneering a systematic, quantitative approach to the design and evaluation of computer architecture with enduring impact on the microprocessor industry.



This book is essential reading for those tasked with leading any complex enterprise in the academic, not-for-profit, or for-profit sector.

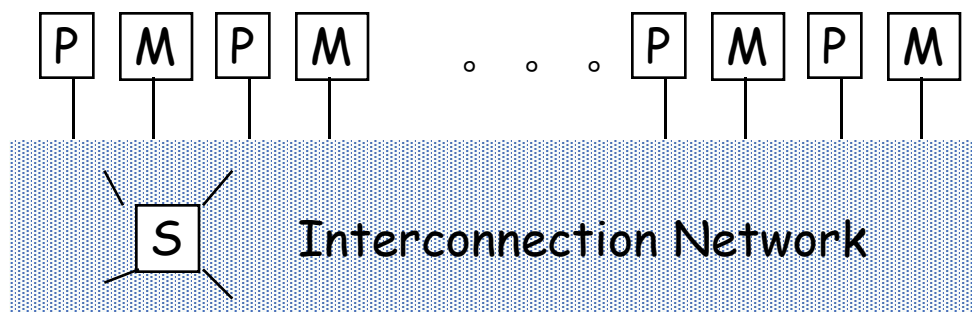


Topics in CA:





Computer Architecture Topics



Processor-Memory-Switch

Multiprocessors
Networks and Interconnections

MCP: Multi-Core Processor
(Chip level Processor)

Shared Memory,
Message Passing,
Data Parallelism

Network Interfaces

Topologies,
Routing,
Bandwidth,
Latency,
Reliability



Topics in this Class (6th. Edition)



	Topic	Reference
Ch1.	Fundamentals of Quantitative Design and Analysis (Quantitative principles)	Chapter 1
Review	Pipelining: Basic and Intermediate Concepts (How to design a pipelined CPU)	Appendix C
Ch2.	Memory Hierarchy Design (How to improve cache and memory)	Chapter 2 Appendix B
Ch3.	Instruction-Level Parallelism and Its Exploitation (How to implement dynamic scheduling)	Chapter 3 Appendix A
Ch4.	Data-Level Parallelism in Vector, SIMD, and GPU Architectures	Chapter 4
Ch5.	Thread-Level Parallelism	Chapter 5



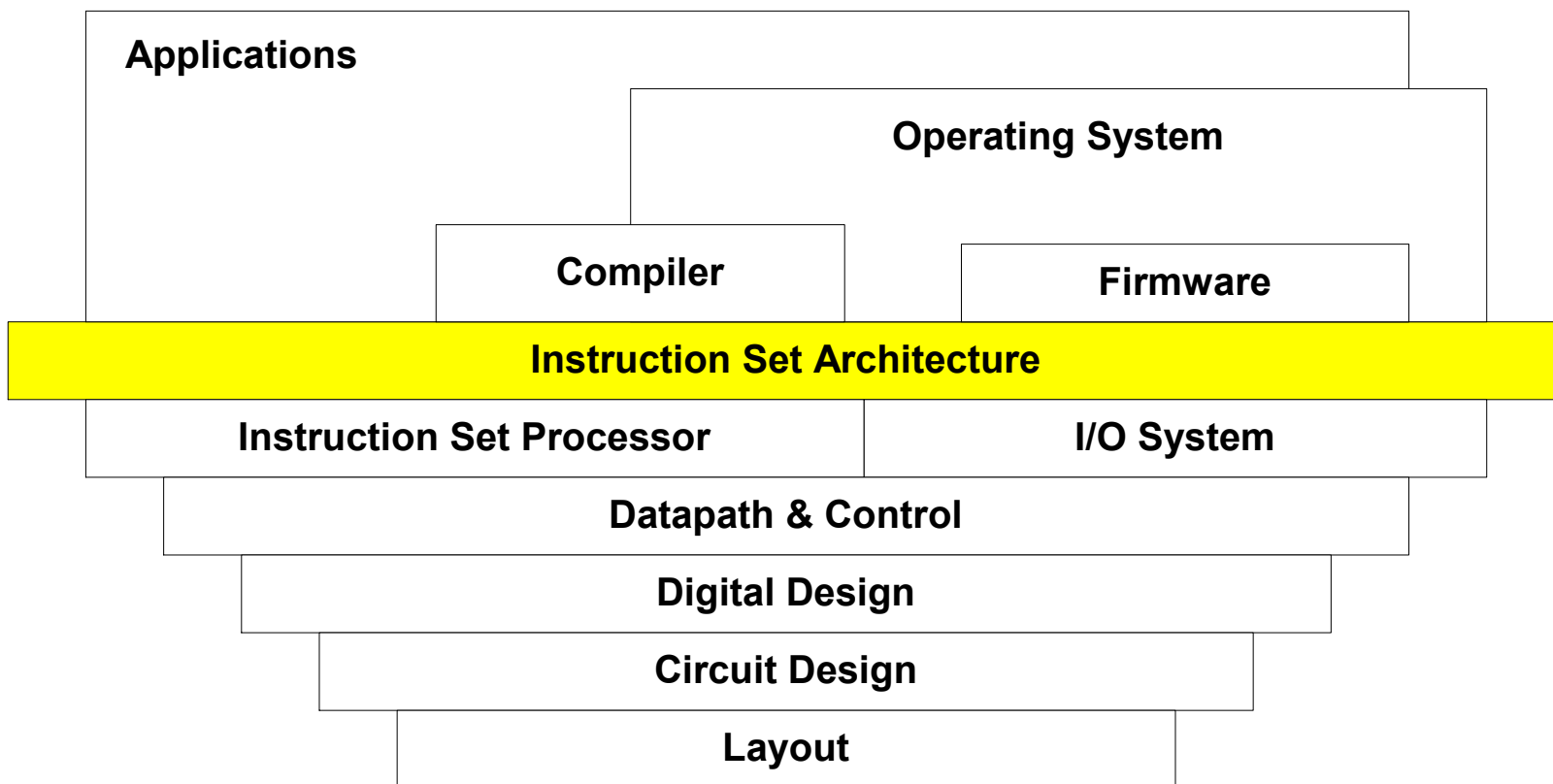
Different from Architecture



■ Part & Whole

■ Bottom up & Top down

■ How & Why





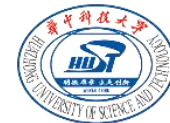
Course Objectives



- The objective of this course
 - Introduce the fundamental techniques on which high-performance computing is based
 - Develop the foundations for analyzing the benefits of design options in computer architecture.
 - systemically learn the fundamental concepts and design approaches of computer architecture from [the view of the whole computer system.](#)
 - Learn the hardware design approaches and to use [hardware design toolkits](#) skillfully.



How ?



- Concepts, Ideas and Principles
- Quantitative approaches
- Hit the problem and right way to solve problem

- *As a man sows, so he shall reap.*

一分耕耘一分收获



What we most care about ?



- For the designer, need to know **why** ?
 - Know the principle & the way to evaluate
 - make decision / make choice.
- For the designer, need to know universal approach
 - Investigate → problem → solve, make decision → Implement → Evaluate



课程成绩与组成



- 作业与考勤: 20%, 每章交一次作业
- 实验/报告: 20% --- 要求撰写实验报告, 不得互相抄袭
实验内容Cache调度
- 期末考试: 60%



- ✓ <https://inst.eecs.berkeley.edu/~cs152>

(UC Berkeley Computer Architecture and Engineering)

- ✓ <https://csdiy.wiki/体系结构/CS61C/>

(UC Berkeley Computer Architecture and Engineering)

- ✓ http://users.ece.cmu.edu/~jhoe/doku/doku.php?id=18447_course_sch edule_spring_2023

(Carnegie Mellon Introduction to Computer Architecture)

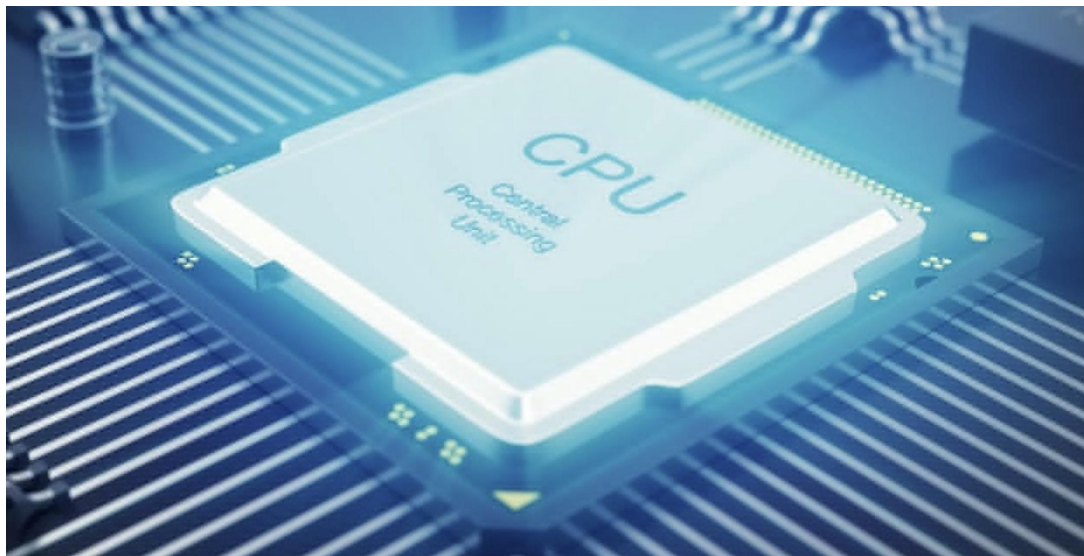
- ✓ <https://safari.ethz.ch/architecture/fall2022/doku.php>

(Eidgenössische Technische Hochschule Zürich)



“The most beautiful thing we can experience is the mysterious. It is the source of all true art and Science.”

---Albert Einstein, What I believe, 1930



保持好奇、保持敬畏、保持对未知的向往，这才是人最珍贵、最美的体验，也是艺术和科学真正的起点。我们要做的就是探索信息世界的神秘。

AI的里程碑：从AlphaGo到ChatGPT



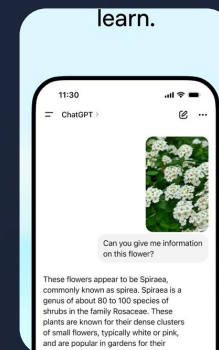
2016年：AlphaGo

击败围棋世界冠军李世石，展示了AI在复杂策略游戏和直觉决策方面的惊人能力，震惊世界。



2017年：Transformer

Google提出Transformer架构，彻底改变了自然语言处理领域，成为后续所有大语言模型的基础。



2022年：ChatGPT

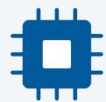
凭借强大的对话和生成能力成为全球现象级应用，标志着生成式AI时代的到来。

技术突破不断重塑人类对人工智能的认知与期待

AI里程碑与计算机系统结构的相互驱动



AI 发展里程碑



早期AI (1950s-1980s)

逻辑推理、专家系统、符号计算 | 基于规则的智能



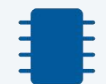
机器学习兴起 (1990s-2010)

统计学习、浅层神经网络 | 从数据中学习模式



深度学习革命 (2012 - AlexNet)

深层神经网络、海量数据 | 特征的自动提取与表示



专用AI芯片时代 (2015s - TPU)

模型规模激增、训练成本高 | 追求极致效率与能效



大模型/生成式AI时代 (2020s - GPT)

超大规模参数、高并发推理 | 通用人工智能探索

计算机系统结构技术变迁



通用CPU与冯·诺依曼架构

串行计算、通用指令集 | AI是通用计算机的普通应用



多核CPU与SIMD优化

多核并行、向量指令 | 推动CPU增加并行机制



异构计算与GPU主导

GPU的SIMT架构、HBM | 引爆异构计算时代



领域专用架构(DSA)与近存计算

专用AI芯片、张量核 | 架构走向专用化和高效能



分布式架构与存算一体

Chiplet、高速互联、存算一体 | 突破存储与带宽墙

核心关系总结：需求与技术的正向循环



AI驱动架构创新 (需求侧)

- 提出新需求：海量计算、高带宽、低功耗
- 架构瓶颈：冯·诺依曼“存储墙”、能效不足
- 技术创新：异构计算、专用芯片、近存计算

架构支撑AI发展 (供给侧)

- 提供新能力：超强并行计算、高吞吐、大容量
- 突破瓶颈：模型规模扩大、训练速度百倍提升
- 进入新时代：从感知智能迈向认知智能

演进历程与关键里程碑

早期阶段

AI作为通用应用，对架构影响较小，依赖通用CPU。

深度学习爆发

引爆异构计算，GPU成为核心，推动并行架构发展。

专用芯片时代

推动领域专用架构(DSA)、近存计算和高带宽存储。

大模型新纪元

驱动分布式架构、Chiplet、存算一体等新技术发展。

总结：AI提出需求，架构通过创新满足需求，二者相互促进，形成持续演进的正向循环。

谢谢大家

